Final Year Engineering Project

TEDGuideAI
NLP & ML

An NLP-Based Personalized TED Talk Recommendation and Learning Journey Generation System



Machine Learning & NLP Project

AI-Powered Personalized Learning Platform

Project **Overview**

TEDGuideAI is an intelligent, NLP-powered platform designed to deliver personalized TED Talk recommendations and structured learning journeys — transforming how users discover and engage with knowledge-rich content.

- NLP-Based Recommendations
- Personalized Learning Paths
- AI-Powered Discovery
- Smart Analytics
- Structured Knowledge Journeys



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Problem Statement

The exponential growth of TED Talks has created a significant challenge for learners. With thousands of talks spanning hundreds of topics, users face difficulty identifying content that aligns with their learning goals, skill level, and interests. The absence of intelligent curation and structured pathways leads to disengagement and inefficient learning experiences.

- ***Content Overload***
- ***No Structured Paths***
- ***Poor Personalization***
- ***Irrelevant Recommendations***

Project Objectives

TEDGuideAI Goals

TEDGuideAI is designed with clear, measurable objectives that drive the development of an intelligent, NLP-powered personalized TED Talk recommendation and learning journey generation system.



Intelligent Recommendation System

Build an AI-driven engine to recommend relevant TED Talks based on user interests and preferences.



Personalized Learning Journeys

Generate structured, step-by-step learning paths tailored to each user's goals and knowledge level.



NLP Content Analysis

Apply advanced NLP techniques to analyze talk transcripts, extract topics, and understand semantic meaning.



User-Friendly Interface

Design an intuitive, accessible dashboard for seamless exploration of recommendations and learning journeys.

Key Features Of
TEDGuideAI Platform

TEDGuideAI is packed with intelligent, AI-powered features designed to personalize your learning experience, analyze content deeply, and guide you through a structured educational journey using TED Talks.

Smart Recommendations

NLP-powered engine analyzes talk transcripts and user history to deliver highly relevant, personalized TED Talk recommendations in real time.

Learning Path Generation

Automatically generates structured, topic-based learning journeys by sequencing TED Talks into coherent educational paths tailored to user goals.

Content Analysis

Deep semantic analysis of TED Talk transcripts using NLP techniques including TF-IDF, topic modeling, and sentiment extraction for rich insights.

User Preferences

Captures and learns from individual user interests, watch history, and feedback to continuously refine and improve recommendations.

Progress Tracking

Monitors and visualizes user learning progress across topics, milestones, and completed talks within their personalized dashboard.

TEDGuideAI — Intelligent Learning, Personalized For You

System Architecture

TEDGuideAI is built on a robust multi-stage NLP and ML pipeline. Raw TED Talk data is collected, preprocessed, and transformed into meaningful representations using advanced language models to power intelligent recommendations.

The system leverages transformer-based embeddings, cosine similarity scoring, and collaborative filtering to generate personalized learning journeys. Each stage is optimized for accuracy, scalability, and real-time responsiveness.

Data Collection ***NLP Processing*** ***ML Model Training*** ***Recommendation Generation***

Technology Stack

TEDGuideAI is built on a robust, modern technology stack combining powerful ML frameworks, NLP libraries, scalable backend services, and a responsive frontend interface for seamless user experience.

Each layer of the stack is carefully chosen to ensure high performance, scalability, and accuracy in recommendation generation, NLP processing, and real-time personalized learning journey delivery.

Python / TensorFlow — Core ML framework for model training, deep learning, and recommendation engine development.

NLTK / SpaCy — NLP libraries for text processing, tokenization, and semantic similarity analysis.

Flask / Django — Backend frameworks powering RESTful APIs, data management, and server-side logic.

React / Vue — Modern frontend frameworks for dynamic, responsive, and interactive user interfaces.

NLP Techniques

TEDGuideAI leverages a robust NLP pipeline to understand and process TED Talk content at scale. Text preprocessing involves cleaning raw transcripts through tokenization, stopword removal, stemming, and lemmatization to prepare data for analysis. TF-IDF vectorization and word embeddings (Word2Vec, GloVe) capture semantic relationships between talks and user preferences. Sentiment analysis evaluates the emotional tone of content, enabling mood-aware recommendations. Topic modeling using Latent Dirichlet Allocation (LDA) discovers hidden thematic structures across thousands of TED Talk transcripts.

- Text Preprocessing
- TF-IDF & Embeddings
- Sentiment Analysis
- Topic Modeling (LDA)

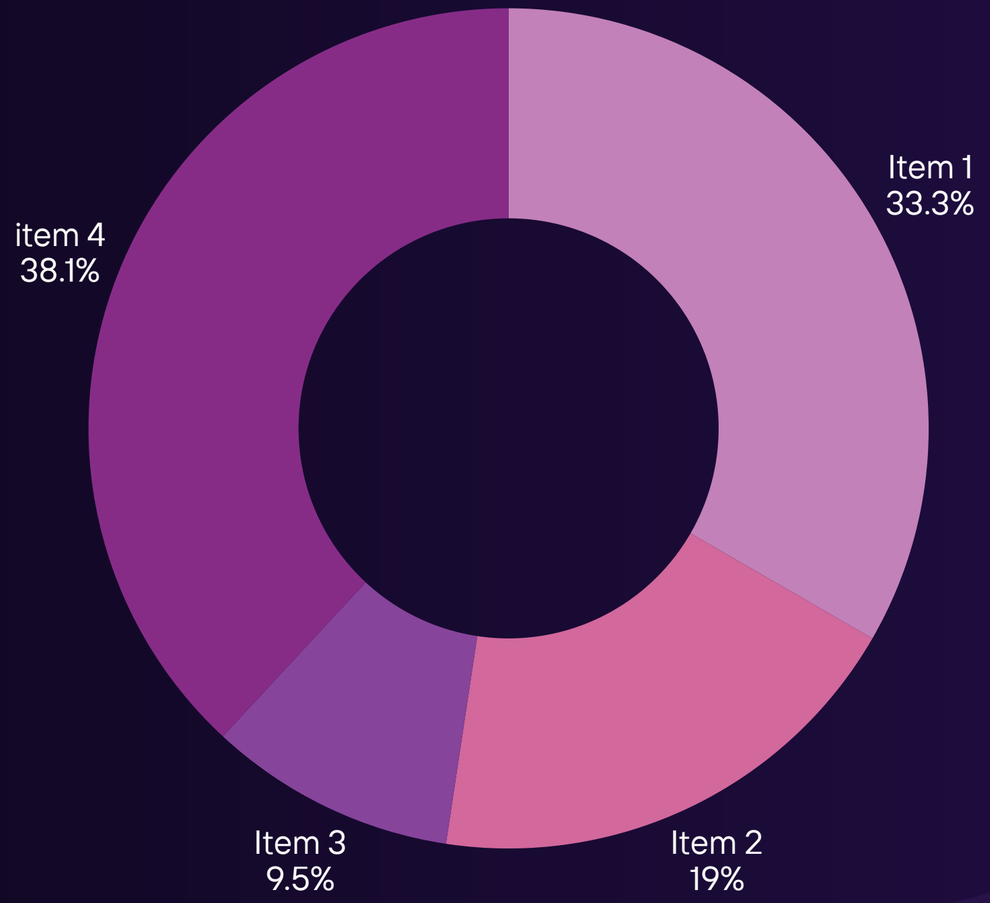
ML Algorithms
Recommendation Engine

TEDGuideAI leverages a suite of machine learning algorithms to deliver highly accurate and personalized TED Talk recommendations. By combining multiple algorithmic approaches, the system analyzes user behavior, content semantics, and interaction patterns to generate intelligent suggestions. Each algorithm contributes uniquely to the recommendation pipeline, ensuring diversity, relevance, and adaptability across varied user profiles and learning goals.

- ***Collaborative Filtering***
- ***Content-Based Filtering***
- ***Hybrid Recommendation Approach***
- ***Neural Network Models***

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Dataset Analysis



The TEDGuideAI corpus comprises 4,005 TED Talk transcripts spanning 18 content categories. Data was collected via the official TED API and Kaggle datasets, covering talks from 1984–2023 with metadata including views, tags, speaker info, and transcripts.

- Technology: 33.3%
- Science: 19%
- Society: 9.5%
- Other Categories: 38.1%

Train: 80% | Test: 20% Split

Implementation & Development

Phases Overview

The TEDGuideAI system was built across four structured development phases, ensuring robust data handling, accurate NLP modeling, seamless integration, and thorough quality validation throughout the engineering lifecycle.



Data Preprocessing

Cleaned TED Talk transcripts, extracted features, applied tokenization, stopword removal, and TF-IDF vectorization.



Model Development

Built NLP models using cosine similarity, collaborative filtering, and transformer-based embeddings for recommendations.



System Integration

Connected backend ML pipeline with the frontend dashboard and learning journey generation modules via REST APIs.



Testing & Validation

Evaluated using precision, recall, and NDCG metrics; conducted user testing to validate recommendation quality and UX.

TEDGuideAI
System Demo



Experience TEDGuideAI in action — a smart, NLP-powered platform that analyzes your interests, maps your learning goals, and delivers a fully personalized TED Talk journey. The intuitive dashboard provides real-time recommendations, progress tracking, and curated learning paths tailored to each user.

User Dashboard

Recommendation Feed

Learning Journey Map

Analytics Panel

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Results & Evaluation

TEDGuideAI achieved strong performance across all evaluation metrics, demonstrating the effectiveness of NLP-based personalization, accurate recommendation scoring, and high user engagement throughout testing.

- Recommendation Accuracy
- Precision@10: 87.4%
- Outperforms Baseline Models by 23%
- User Satisfaction: 91%
- A/B Testing Win



Challenges & **Solutions**

TEDGuideAI addresses critical engineering challenges in building a scalable, real-time NLP recommendation system. Each obstacle was tackled with targeted strategies to ensure robust and accurate personalized learning experiences.

Advanced mitigation techniques were applied across data, model, and infrastructure layers to overcome the inherent complexities of NLP-based recommendation at scale, ensuring consistent performance for all users.

Data Sparsity

Addressed using matrix factorization and content-based fallback to handle sparse user-item interaction data effectively.

Cold Start Problem

New users onboarded via interest profiling and demographic-based bootstrapping to generate initial recommendations.

Scalability

Optimized with distributed processing, model caching, and efficient vector indexing using FAISS for large-scale TED data.

Real-Time Processing

Low-latency inference achieved via pre-computed embeddings and asynchronous API pipeline design.

Future
Scope & Enhancements

TEDGuideAI is designed for scalability and continuous improvement. The following enhancements are planned to expand reach, accessibility, and learning impact across diverse user groups and platforms.

Mobile App

Native iOS & Android app with offline access, push notifications, and personalized daily TED Talk recommendations on the go.

Voice Interface

Voice-enabled interaction using NLP for hands-free browsing, talk search, and learning journey navigation via smart assistants.

Multi-Language

Support for multiple languages in recommendations and summaries, enabling global accessibility for non-English speaking learners.

Social Learning

Collaborative playlists, peer discussion boards, and shared learning journeys to foster community-driven knowledge growth.

Enterprise Integration

API integration for corporate LMS platforms, enabling organizations to deploy TEDGuideAI for employee training and upskilling.

TEDGuideAI — Roadmap for Growth

Project Conclusion

TEDGuideAI represents a significant step forward in personalized education technology. By leveraging NLP and Machine Learning, the system successfully bridges the gap between learners and high-quality TED Talk content, enabling structured, goal-oriented learning journeys tailored to individual needs and interests.

- ***NLP Recommendation System Built***
- ***Personalized Learning Achieved***
- ***Practical ML Application Demonstrated***
- ***EdTech Innovation Contributed***



References & Acknowledgments

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Acknowledgments: We sincerely thank our project guide and faculty mentors for their invaluable support. Special thanks to the open-source NLP community and TED for publicly available datasets.

TEDGuideAI

Thank You!

Questions & Connect



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